08 The Application Layer

08.05 – 08.06 CDN - Content Delivery Network, CDN e Video Streaming

Content in today's Internet

- Most flows are HTTP
  - Web is at least 52% of traffic
  - Median object size is 2.7K, average is 85K (as of 2007)
- HTTP uses TCP, so it will
  - Be ACK clocked
  - For Web, likely never leave slow start
- Is the Internet designed for this common case?
  - Why?

Evolution of Serving Web Content

- In the beginning…
  - …there was a single server
  - Probably located in a closet
  - And it probably served blinking text
- Issues with this model
  - Site reliability
    - Unplugging cable, hardware failure, natural disaster
  - Scalability
    - Flash crowds (aka Slashdotting)

Replicated Web service

- Use multiple servers
- Advantages
  - Better scalability
  - Better reliability
- Disadvantages
  - How do you decide which server to use?
  - How to do synchronize state among servers?

Load Balancers

- Device that multiplexes requests across a collection of servers
  - All servers share one public IP
  - Balancer transparently directs requests to different servers

- How should the balancer assign clients to servers?
  - Random / round-robin
    - When is this a good idea?
  - Load-based
    - When might this fail?
- Challenges
  - Scalability (must support traffic for n hosts)
  - State (must keep track of previous decisions)
    - RESTful APIs reduce this limitation

Load balancing: Are we done?

- Advantages
  - Allows scaling of hardware independent of IPs
  - Relatively easy to maintain
- Disadvantages
  - Expensive
  - Still a single point of failure
  - Location!

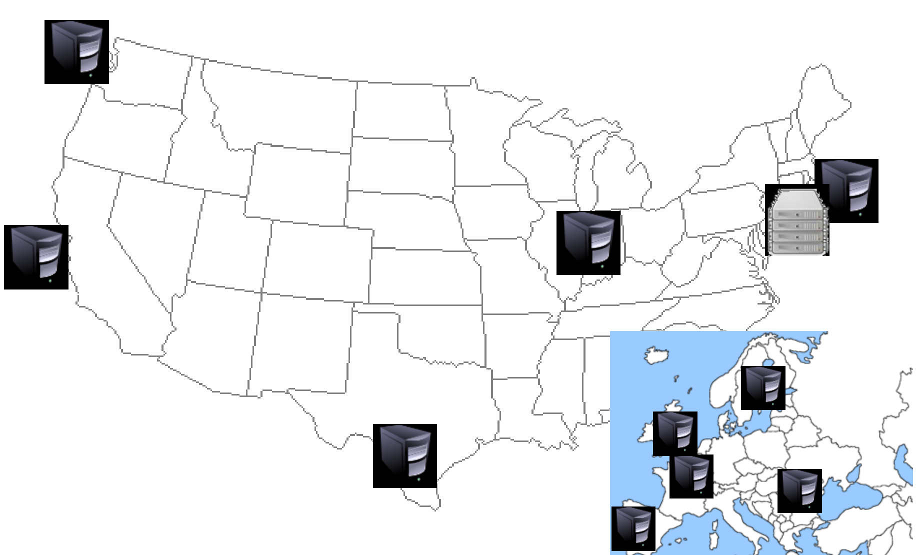
Where do we place the load balancer for Wikipedia?

Popping up: HTTP performance

- For Web pages
  - RTT matters most
  - Where should the server go?
- For video
  - Available bandwidth matters most
  - Where should the server go?

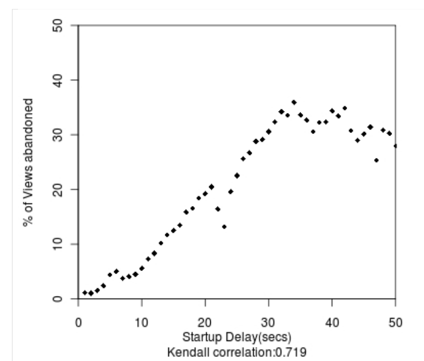
- Is there one location that is best for everyone?

Server placement



Why speed matters

- Impact on user experience
  - Users navigating away from pages
  - Video startup delay



Why speed matters

- Impact on user experience
  - Users navigating away from pages
  - Video startup delay
- Impact on revenue
  - Amazon: increased revenue 1% for every 100ms reduction in PLT (page load time)
  - Shopzilla:12% increase in revenue by reducing PLT from 6 seconds to 1.2 seconds

Strawman solution: Web caches

- ISP uses a middlebox that caches Web content
  - Better performance – content is closer to users
  - Lower cost – content traverses network boundary once
  - Does this solve the problem?
- No!
  - Size of all Web content is too large
    - Zipf distribution limits cache hit rate
  - Web content is **dynamic** and **customized**
    - Can't cache banking content
    - What does it mean to cache search results?

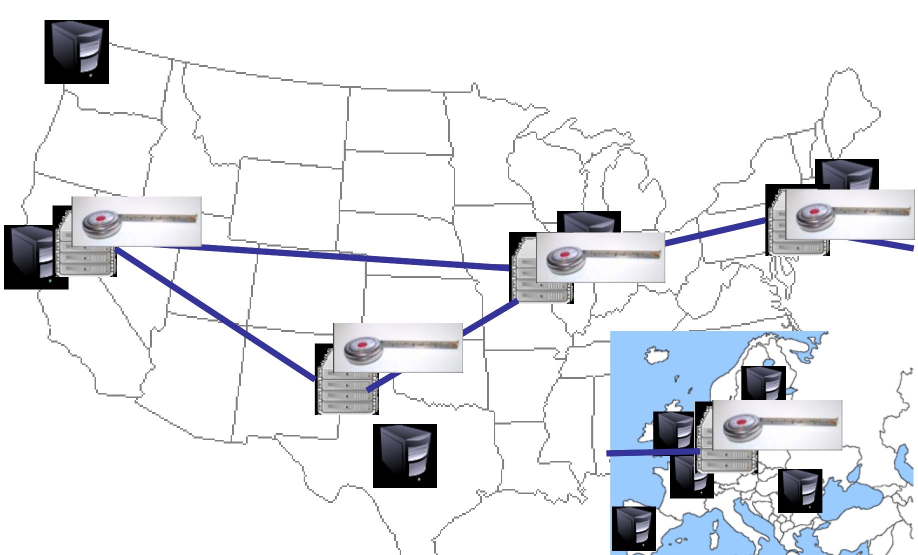
What is a CDN?

- Content Delivery Network
  - Also sometimes called Content Distribution Network
  - At least half of the world's bits are delivered by a CDN
    - Probably closer to 80/90%
- Primary Goals
  - Create replicas of content throughout the Internet
  - Ensure that replicas are always available
  - Directly clients to replicas that will give good performance

Key Components of a CDN

- Distributed servers
  - Usually located inside of other ISPs
  - Often located in IXPs
  - High-speed network connecting them
- Clients (eyeballs)
  - Can be located anywhere in the world
  - They want fast Web performance
- Glue
  - Something that binds clients to “nearby” replica servers

Key CDN Components



Examples of CDNs

- Akamai
  - 147K+ servers, 1200+ networks, 650+ cities, 92 countries
- Limelight
  - Well provisioned delivery centers, interconnected via a private fiber-optic connected to 700+ access networks
- Edgecast
  - 30+ PoPs, 5 continents, 2000+ direct connections
- Others
  - Google, Facebook, AWS, AT&T, Level3, Brokers

Inside a CDN

- Servers are deployed in clusters for reliability
  - Some may be offline
    - Could be due to failure
    - Also could be “suspended” (e.g., to save power or for upgrade)
- Could be multiple clusters per location (e.g., in multiple racks)
- Server locations
  - Well-connected points of presence (PoPs)
  - Inside of ISPs

Mapping clients to servers

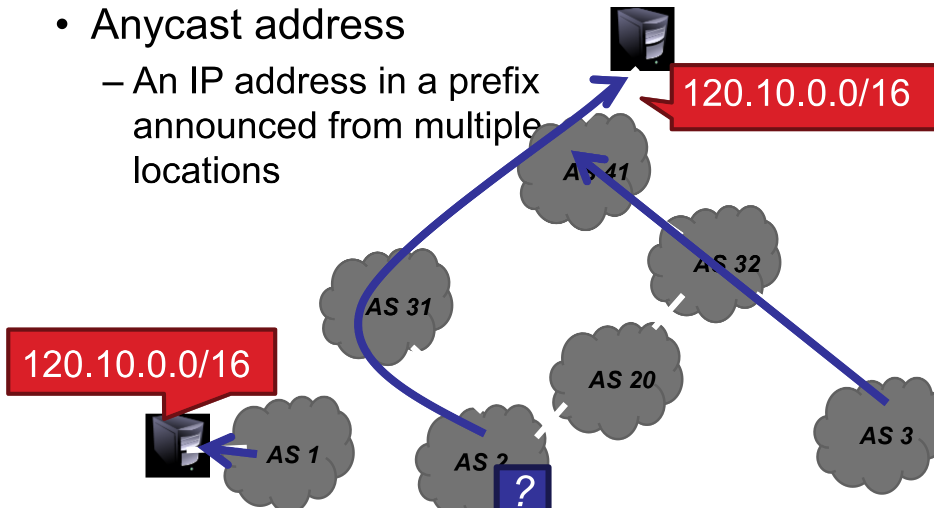
- CDNs need a way to send clients to the “best” server
  - The best server can change over time
  - And this depends on client location, network conditions, server load, …
  - What existing technology can we use for this?
- DNS-based redirection
  - Clients request www.foo.com
  - DNS server directs client to one or more IPs based on request IP
  - Use short TTL to limit the effect of caching

DNS Redirection Considerations

- Advantages
  - Uses existing, scalable DNS infrastructure
  - URLs can stay essentially the same
  - TTLs can control “freshness”
- Limitations
  - DNS servers see only the DNS server IP
    - Assumes that client and DNS server are close. Is this accurate?
  - Small TTLs are often ignored
  - Content owner must give up control
  - Unicast addresses can limit reliability

CDN Using Anycast

- Anycast address
  - An IP address in a prefix announced from multiple locations



How does Anycast Work?

- Anycast network routing is able to route incoming connection requests across multiple data centers.
- When requests come into a single IP address associated with the Anycast network, the network distributes the data based on some prioritization methodology.
- The selection process behind choosing a particular data center will typically be optimized to reduce latency by selecting the data center with the shortest distance from the requester.
- Anycast is one of the 5 main network protocol methods used in the Internet protocol.

Anycasting Considerations

- Why do anycast?
  - Simplifies network management
    - Replica servers can be in the same network domain
  - Uses best BGP path
- Disadvantages
  - BGP path may not be optimal
  - Stateful services can be complicated

How does an Anycast network mitigate a DDoS attack?

Optimizing Performance

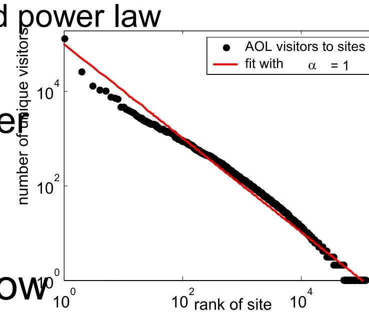
Key goal

Send clients to server with best end-to-end performance

- Performance depends on
  - Server load
  - Content at that server
  - Network conditions
- Optimizing for server load
  - Load balancing, monitoring at servers
  - Generally solved

Optimizing performance: caching

- Where to cache content?
  - Popularity of Web objects is Zipf-like
    - Also called heavy-tailed and power law
  - $N_r \sim r^{-1}$
  - Small number of sites cover large fraction of requests



- Given this observation, how should cache-replacement work?

Optimizing performance: Network

- There are good solutions to server load and content
  - What about network performance?
- Key challenges for network performance
  - Measuring paths is hard
    - Traceroute gives us only the forward path
    - Shortest path != best path
  - RTT estimation is hard
    - Variable network conditions
    - May not represent end-to-end performance
  - No access to client-perceived performance

Optimizing performance: Network

- Example approximation strategies
  - Geographic mapping
    - Hard to map IP to location
    - Internet paths do not take shortest distance
  - Active measurement
    - Ping from all replicas to all routable prefixes
    - 56B * 100 servers * 500k prefixes = 500+MB of traffic per round
  - Passive measurement
    - Send fraction of clients to different servers, observe performance
    - Downside: Some clients get bad performance

Streaming Content Delivery Networks

- The recent exponential increase in video streaming has caused the major telecommunications companies to become CDNs in order to exploit their huge capital investments in video media and to retain subscribers.
- Telcos like Verizon, AT&T and Sprint offer varying levels of services to streaming broadcasters. Obviously, these firms own their own continental networks with “into the home” accessibility, so certain cost advantages may be realized.

Streaming Content Delivery Networks

- Most streaming video equipment has a method by which you can select the content delivery network, which will carry your program to audiences worldwide.
- Typically, you download a “profile” from a CDN and then configure that profile on your system.
- The profile allows for a running account of your bill and other necessary information that you provide, such as location, resolution and audio “headroom.”

Large CDN

- Competing with the telcos by installing their own PoPs and nodes are large, traditional CDNs like
  - Akamai, which streams video for the Canadian Broadcasting Corporation, AvidMobile and MTV;
  - Level 3, which serves government, healthcare and financial firms as well as the network video distribution of the Fox Network;
  - Limelight Networks, which caters to the video gaming needs of Nintendo and the business-to-business (B2B) videos needs of large corporate clients.

Video Optimized CDNs

- By focusing on video-specific needs, these CDNs offer some features that are unique to video streaming, such as
  - video social media integration;
  - audience analytics;
  - multi-format transcoding for mobile and tablets;
  - over-the-top (OTT) distribution;
  - monetization (collecting fees to view your videos);
  - attention to 2K and 4K high definition formats.

Video Optimized CDNs

- One of the best-known video optimized CDN is **Ustream**, which offers separate product services for what they consider to be the three dominant categories of video streaming: Pro Broadcasting, Corporate Internal Communications (Ustream Align) and lead-generation marketing (Ustream Demand).
- Another video optimized CDN, **JET-Stream** (former **Streamzilla)**, is gaining share in Europe and may be a good local choice if your audience is on this side of the Atlantic. In addition to streaming, Streamzilla also offers resellers the kinds of tools they need to generate revenue and create a network presence.

Jet-Stream from StreamZilla

- StreamZilla became the basis for a new idea.
- The new approach is: an implemented HTTP adaptive bit rate streaming to go beyond what regular CDNs offer. HTTP streaming opened up new opportunities.
- Jet-stream redefines who gets traffic, and business, integrating all major CDNs in the platform.
- Then some new load balancing algorithms: they proved they could reduce streaming costs by 50%, much higher uptime and guaranteed performance!

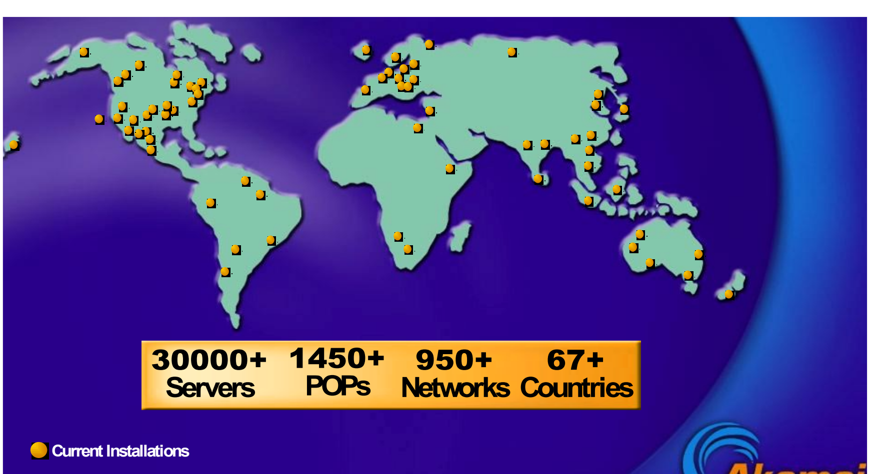
MultiCDN

- Some of new capabilities are:
  - Enforced load balancing: it instantly load balance traffic, without any dependency on third party DNS systems (if one CDN fails, we instantly kick them out of the pool)
  - CDN stacking: it load balance over a pool of CDNs, effectively building the largest CDN in the world (use the capacity of all CDNs together)
  - Fine granular load balancing: optimised load balancing per ISP, down to IP addresses
  - Deep CDN integration: control global CDNs as if they are our own edge servers: flushing, monitoring, access control, reporting, log file analytics, all automated
  - Mix Global CDNs and edge servers: optimise traffic over a hybrid mix of resources: your own CDN and your own edges, plus global CDNs

Akamai case study

- Deployment
  - 147K+ servers, 1200+ networks, 650+ cities, 92 countries
  - highly hierarchical, caching depends on popularity
  - 4 yr depreciation of servers
  - Many servers inside ISPs, who are thrilled to have them
  - Deployed inside100 new networks in last few years
- Customers
  - 250K+ domains: all top 60 eCommerce sites, all top 30 M&E companies, 9 of 10 top banks, 13 of top 15 auto manufacturers
- Overall stats
  - 5+ terabits/second, 30+ million hits/second, 2+ trillion deliveries/day, 100+ PB/day, 10+ million concurrent streams
  - 15-30% of Web traffic

Somewhat old network map



Akamizing Links

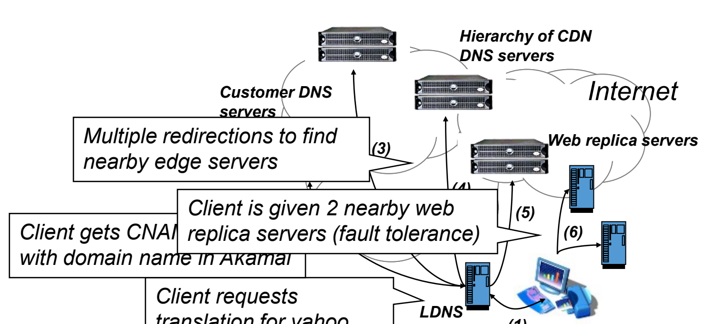
- Embedded URLs are Converted to ARLs

AK

```
<html>
<head>
<title>Welcome to xyz.com!</title>
</head>
<body>
<img src="http://www.xyz.com/logos/logo.gif">
<img src="http://www.xyz.com/jpgs/navbar1.jpg">
<h1>Welcome to our Web site!</h1>
<a href="page2.html">Click here to enter</a> </body>
</html>
```


DNS Redirection

- Web client's request redirected to 'close' by server
  - Client gets web site's DNS CNAME entry with domain name in CDN network
  - Hierarchy of CDN's DNS servers direct client to 2 nearby servers



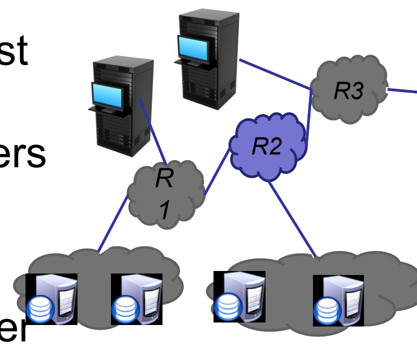
Mapping Clients to Servers

- Maps IP address of client's name server and type of content being requested (e.g., "g" in a212.g.akamai.net) to an Akamai cluster.
- Special cases: Akamai Accelerated Network Partners (AANPs)
  - Probably uses internal network paths
  - Also may require special "compute" nodes
- General case: "Core Point" analysis

Core points

Core points

- Core point is the first router at which all paths to nameservers intersect



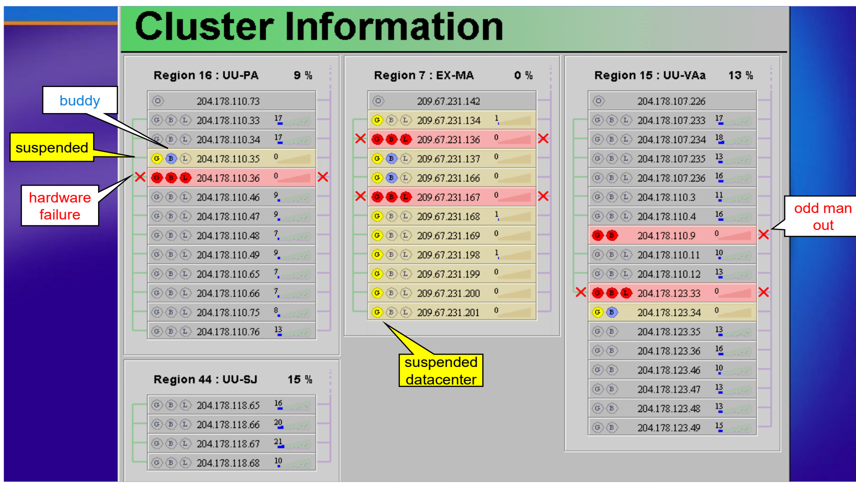
- Traceroute once per day from 300 clusters to
  280,000 core points

Core Points

- 280,000 nameservers (98.8% of requests) reduced to 30,000 core points
  – Nice order-of-magnitude reduction
- ping core points every 6 minutes
  – More than 200 probes per second per server!

- Note the use of low-rate expensive measurements with high-rate cheap measurements

Server clusters



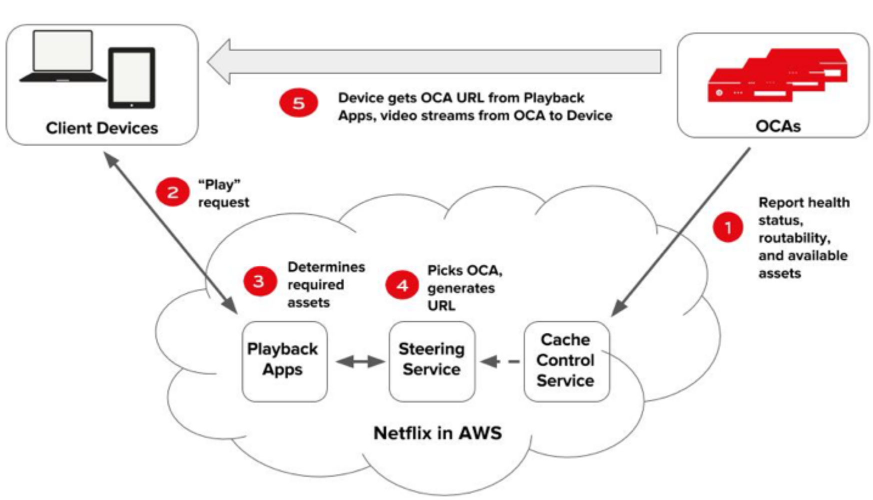
Key future challenges

- Mobile networks
  - Latency in cell networks is higher
  - Internal network structure is more opaque
- Video
  - 4k/8k UHD = 16-30K Kbps compressed
  - 25K Tbps projected
  - Big data center networks not enough (5 Tbps each)
  - Multicast (from end systems) potential solution

Netflix case study

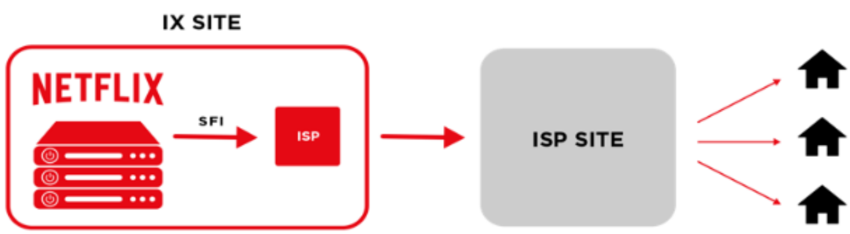
- Open Connect is the name of the global network that is responsible for delivering Netflix TV shows and movies worldwide, and it's a CDN.
- The building blocks of Open Connect are suite of purpose-built server appliances, called Open Connect Appliances (OCAs).
- These appliances store and serve our video content, with the sole responsibility of delivering playable bits to client devices as fast as possible.

How Netflix works



Types of deployments - IX

- Netflix install OCAs within IXPs in significant Netflix markets throughout the world. These OCAs are interconnected with mutually - present ISPs via settlement free public or private peering (SFI).



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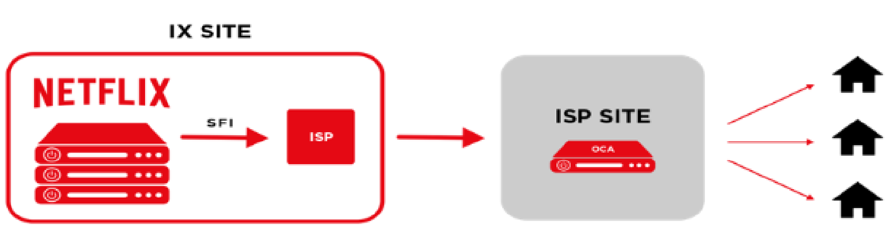
Types of deployments - Embedded

- Netflix provides OCAs free of charge to qualifying ISPs. These OCAs, with the same capabilities as the OCAs in the IXPs, are deployed directly inside ISP networks.
- Netflix provides the server hardware and the ISPs provide power, space, and connectivity. ISPs directly control which of their customers are routed to their embedded OCAs.



Traffic delivery from an OCA

- The traffic directed to the appliance is determined explicitly by the ISP partner and by Netflix, not by the appliance itself.
- An OCA only serves clients at IP addresses that are advertised to the OCA via a BGP session.
- If content is requested that is not contained on an embedded OCA, the client request is directed to the closest Netflix content site via peering (if present) or via transit.
- Netflix manages clients to its OCAs based on an ISP's BGP advertisements, coupled with the routing and steering algorithms in the Open Connect control plane.

Appliance selection criteria

1. Longest Match — The appliance that receives the most-specific route to the client's prefix.
2. Shortest AS path — The appliance that receives the route to the client's netblock with the shortest AS path.
3. Lowest MED — The appliance that receives the route to the client's netblock with the lowest multi-exit discriminator (MED).
4. Geolocation — We geolocate based on client IPs, comparing it to the latitude and longitude of nearby OCAs to determine the closest available system.